Lab 06 - Balance Reliability, Security and Delivery Guardrails

Course: DevOps Foundation Training (TGS-2024049184) Version: v4.0 Topic: 06 Estimated time: 35-45 minutes

Workplace scenario

A team wants faster deployment, while operations and security want more approval gates. You must use risk and error-budget evidence to design a faster but controlled path.

Objective

Apply K4 and A4 by categorising reliability and security risks, then selecting proportional automated guardrails.

What you will produce

A guardrail_decisions.csv, error_budget.png and documented trade-off decision.

Prerequisites

- R 4.x available as Rscript.
- Work from this lab folder so relative paths resolve.
- Use only the supplied synthetic data; no internet connection or external package is required.
- Preserve the source data. Make experimental changes in a copied file, then restore it.

Folder map

```
Lab-06-reliability-security-guardrails/
■■■ README.md / README.pdf
■■■ CHECKLIST.md / CHECKLIST.pdf
■■■ reliability_security_guardrails.R
■■■ data/
■■■ output/            # reproducible evidence produced by the R script
```

Detailed procedure

Step 1 - Inspect service_risk.csv and review the SLO, observed reliability and security finding fields.

```
Rscript -e 'print(read.csv("data/service_risk.csv"))'
```

Record what you observed before continuing.

Step 2 - Run resilience_security.R to calculate error-budget consumption and security risk score.

```
Rscript resilience_security.R
```

Record what you observed before continuing.

Step 3 - Review guardrail_decisions.csv and explain why different services receive warn, approve or block treatment.

```
Rscript -e 'print(read.csv("output/guardrail_decisions.csv"))'
```

Record what you observed before continuing.

Step 4 - Open error_budget.png and identify the service consuming more than its available budget.

```
open output/error_budget.png
```

Record what you observed before continuing.

Step 5 - Propose one preventive, one detective and one recovery control for the highest-risk service.

```
open CHECKLIST.md
```

Record what you observed before continuing.

Step 6 - Record the control owner, evidence source and safe rollback method.

```
open CHECKLIST.md
```

Record what you observed before continuing.

Evidence to retain

[ ] output/guardrail_decisions.csv

[ ] output/error_budget.png

[ ] completed CHECKLIST.md

Acceptance criteria

- Error budget formula correct
- Three distinct proportional policies
- Prevent/detect/recover controls
- Rollback and owner recorded

Verification

All policies are derived from published thresholds; at least one service is blocked, one warns and one proceeds automatically.

Troubleshooting

- cannot open file: confirm your terminal is in this lab folder.
- Rscript: command not found: install R, reopen the terminal and run Rscript --version.

- A validation stop(...) message is evidence that the input broke a stated rule; inspect the named field rather than bypassing the check.
- If an output looks wrong, compare row counts and IDs with the source CSV before changing the analysis.

Clean rerun

Delete only the files inside output/, then run the R script again. The source CSV and script remain unchanged.